



OPEN Characterization and comparative analysis of urinary bacterial microbiome profiling in healthy cats and cats with feline idiopathic cystitis

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Over the past decade, the make up of urinary microbiota and its association with disease has garnered substantial attention. However, the urinary microbiota patterns in healthy cats remain unclear, as do the potential dysbiosis conditions in cats with feline idiopathic cystitis (FIC). We hypothesized that urinary microbiota differ by age and sex in healthy cats, and vary between healthy cats and those with FIC. Using 16S rRNA gene sequencing, we analyzed the urinary microbiota of 25 healthy cats and 15 cats with FIC. Our results indicate that the feline urinary microbiota is primarily composed of Proteobacteria. While no significant differences in bacterial abundance were found between sexes in healthy cats, some taxa showed variations by age. Cats with FIC demonstrated microbiota disruption, characterized by an increase in Firmicutes and Bacilli, along with a decrease in Proteobacteria, particularly a reduction in the *Pseudomonas* genus, compared to healthy cats. These results demonstrate the presence of a bacterial microbiome in the urine of cats and suggest a potential association between altered microbiota and FIC. Multi-omics approaches could provide further insights into the core urobiome and its role in feline urinary tract disease.

Keywords Urinary microbiota, Cat, Feline idiopathic cystitis, Dysbiosis

The first human microbiome research initiative, known as the “Human Microbiome Project,” commenced in 2007 with the goal of investigating and characterizing the human microbiome and analyzing its role in health and disease¹. Subsequently, researchers began exploring the microbiome of the human urinary tract, challenging the conventional notion that urine is sterile. Extensive studies have been conducted to examine the connection between the urinary microbiome and various urinary tract diseases, including bladder cancer², urgency urinary incontinence or overactive bladder³, urolithiasis⁴, urinary tract infections⁵, chronic kidney disease⁶, and interstitial cystitis⁷. Research findings have prompted urologists to consider urinary dysbiosis as a potential factor in various urinary tract disorders.

Similar to humans, microbiome research is expanding within the field of veterinary medicine. However, studies specifically focusing on feline urinary microbiomes remain limited. In 2020, Balboni et al. (2020), reported that no viable bacteria were cultured from the urine of healthy cats; however, 16S rRNA gene sequencing detected trace amounts of bacterial DNA, identified as originating from contaminants⁸. In addition, they further confirmed the absence of viable bacteria in the urine of cats with feline idiopathic cystitis (FIC)⁹. In contrast, Kim et al. (2021) identified distinct microbial communities in the urine of cats that differ from those found in dogs and humans. They also detected predominant bacteria associated with urinary tract infections and chronic kidney disease, suggesting dysbiosis of the urinary microbiome in these affected cats¹⁰. Additionally, two recent studies discovered significant changes in the urinary microbiota of cats with urethral obstruction or kidney stones compared to healthy cats^{11,12}. These findings highlight the need for further investigation into the discrepancies observed in urinary microbiota among cats.

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Studying the urinary microbiomes of healthy cats is essential for establishing a comprehensive understanding of the baseline microbial communities that contribute to feline urinary health. In healthy humans, the urinary microbiome has been shown to vary based on natural factors, including age^{13,14} and gender¹⁴, highlighting the importance of understanding these variations to elucidate how age and sex influence urinary microbiota. Previous studies on the urinary microbiomes of cats have not thoroughly investigated age and gender factors^{8,10,12}. Therefore, exploring the urinary microbiome in healthy cats of varying ages and sexes is warranted. Characterizing the normal urinary microbiome can provide valuable reference information for identifying deviations associated with urinary disorders.

Feline idiopathic cystitis (FIC) is the most common cause of feline lower urinary tract disease (FLUTD), particularly in adult male cats, who are at an increased risk of developing the condition^{15–17}. FIC continues to present challenges for veterinary healthcare systems due to its unclear etiology and limited treatment options. Notably, FIC shares similarities with interstitial cystitis in humans; both conditions involve bladder inflammation and present urinary symptoms, yet their underlying pathophysiology remains poorly understood and is likely influenced by multiple contributing factors¹⁸. Interestingly, variations in urinary microbiota have been reported in humans with interstitial cystitis, where distinct changes in bacterial abundance, particularly a reduction in *Lactobacillus*, were observed compared to healthy individuals^{19,20}. Additionally, the abundance of *Lactobacillus* is higher in the urine of female patients and healthy controls compared to males, while bacterial genera such as *Bacteroides* and *Acinetobacter* are less abundant in females²¹. However, the precise relationship between *Lactobacillus* and the host remains unclear. In the veterinary field, only two studies have explored urinary microbiomes in relation to FIC^{9,22}. Given these limited findings, further investigation of the urinary microbiome could help identify potential microbial imbalances, patterns associated with the disease, and specific microbial signatures that may serve as novel diagnostic biomarkers for FIC.

Therefore, our study aims to investigate and characterize the microbial communities present in the urine of healthy cats, examining the influences of sex and age on the urinary microbiome. Additionally, we aim to compare the urinary microbiota of healthy cats with that of cats diagnosed with idiopathic cystitis. We hypothesize that there will be significant differences in the abundance and diversity of urinary microbiota between male and female cats, as well as between young and adult healthy cats. Furthermore, we anticipate that the abundance and diversity of urinary microbiota will differ markedly between healthy cats and those with idiopathic cystitis. The findings could provide valuable insights into the role of microbial communities in feline urinary health and disease.

Results

Study population

We collected samples for age and sex comparison from a total of 20 healthy cats, all of which were presented for wellness examinations and confirmed to be clinically healthy. The cats included Domestic Shorthair ($n=18$), American Shorthair ($n=1$), and Persian ($n=1$). Of these 20 cats, 10 were females (4 spayed, 6 intact) and 10 were males (4 castrated, 6 intact). The median ages for males and females were 1.1 (IQR = 0.6–2.5) years and 1.0 (IQR = 0.6–2.8) years, respectively, with mean body weights of 3.5 ± 0.6 kg for males and 3.6 ± 0.4 kg for females. When classified into young ($n=10$) and adult ($n=10$) groups, the median ages were 0.6 (IQR = 0.6–0.7) years for young cats and 2.6 (IQR = 1.5–2.9) years for adult cats, with mean body weights of 3.2 ± 0.3 kg for young cats and 3.9 ± 0.4 kg for adult cats (see supplementary Table 1).

For the comparison of the urinary microbiome between cats with FIC and healthy controls, 25 cats (15 with FIC and 10 healthy) were included. Since adult male cats have a higher risk of developing FIC and make up the majority of the FIC population, all cats included in the comparison were male and over 1 year of age. The cats included Domestic Shorthair ($n=14$), Persian ($n=8$), Siamese ($n=2$), and Scottish Fold ($n=1$). Among these cats, 6 were castrated, while the remaining 19 were intact. The mean body weight was 4.8 ± 0.9 kg for the FIC group and 4.4 ± 0.7 kg for the healthy group. The median ages were 3.0 (IQR = 2–5) years for the FIC group and 2.0 (IQR = 1–2) years for the healthy group (see supplementary Table 2).

Sequence analysis

All 20 samples from healthy cats generated a total of 518,785 reads, with an average of 25,939 reads per sample (see Supplementary Table 3). A total of 466 amplicon sequence variants (ASVs) were identified, and they were assigned to 3 phyla, 6 classes, 10 orders, 13 families, and 13 genera.

In the 25 samples from cats with FIC ($n=15$) and healthy controls ($n=10$), a total of 517,782 reads were generated, with an average of 20,711 reads per sample (see Supplementary Table 4). Across all 25 samples, 525 ASVs were identified, assigned to 3 phyla, 5 classes, 10 orders, 12 families, and 12 genera.

Comparative analysis of urinary microbiota diversity in healthy cats by age and sex

We characterized and compared the urinary microbiota of healthy cats by age (young vs. adult) and sex (male vs. female). Bacterial diversity within each sample was assessed using the number of observed species, Chao1, Shannon, and Simpson indices. Young cats exhibited significantly higher diversity than adult cats based on the Shannon and Simpson indices at the class ($p=0.04$), order ($p=0.04$), family ($p=0.01$), and genus ($p=0.01$) levels (see Supplementary Table 5). However, no statistically significant differences were observed between males and females for any of the alpha diversity indices (see Supplementary Table 6).

Beta diversity was analyzed to compare the urinary microbiota composition between groups, using the Bray-Curtis and Jaccard indices. Comparisons between age groups revealed no significant differences between young and adult cats, and a lack of clustering was observed in the principal coordinate analysis (PCoA) plots (see Supplementary Table 7). Additionally, when considering differences between male and female cats, no

statistically significant differences in bacterial structure were identified. Clustering was not evidently observed in the PCoA plot (see Supplementary Table 8).

Comparative analysis of urinary Microbiome composition in healthy cats by age and sex

Taxonomic analysis was conducted to compare the bacterial composition of the urinary microbiota in healthy cats across age and sex groups. In the comparison of age groups, a total of 3 phyla, 6 classes, 10 orders, 13 families, and 13 genera were identified (see Supplementary Table 9, Fig. 1). Notably, Proteobacteria were the most abundant phylum in both young and adult cats, accounting for 90% and 95.6% of the total bacterial composition, respectively. Class Gammaproteobacteria, Order Pseudomonadales, Family *Pseudomonadaceae*, and Genus *Pseudomonas* were predominantly detected in both age groups, constituting up to 50% of each taxonomic level. When comparing young and adult cats, these predominant taxa were found to be significantly lower in young cats than in adults ($p \leq 0.04$). Additionally, several taxa, including Phylum Firmicutes, Class Bacilli, Order Bacillales, and Family *Staphylococcaceae*, were significantly higher abundance in young cats ($p \leq 0.03$). While these taxa were more prevalent in the young cat group, they were not universally present, being detected in only 4 out of 10 individuals. It is important to note that the small sample size ($n = 10$ per group) limits the generalizability of these findings and should be considered when interpreting age-related results.

In the comparison of sex groups, a total of 3 phyla, 6 classes, 10 orders, 13 families, and 13 genera were identified in the bacterial composition. Proteobacteria exhibited the highest proportions in both male and female cats, accounting for up to 90% of the total relative abundance (see Supplementary Table 10, Fig. 2). Additionally, the bacterial composition was prominently characterized by Class Gammaproteobacteria, Order Pseudomonadales, Family *Pseudomonadaceae*, and Genus *Pseudomonas*, which dominated in both sex groups, constituting up to 50% of the total taxa at each taxonomic level. Comparative analysis between male and female cats revealed no statistically significant differences in the relative abundance of bacterial taxa across phylum, class, order, family, and genus levels.

Comparative analysis of urinary microbiota diversity in healthy cats versus cats with feline idiopathic cystitis

When comparing bacterial diversity within samples from cats with idiopathic cystitis and clinically healthy cats, no statistically significant differences were observed in any of the alpha diversity indices (see Supplementary Table 11). However, both the Bray-Curtis and Jaccard indices revealed significant differences in microbial communities between cats with FIC and healthy cats at the phylum, family, and genus levels, as illustrated in the PCoA plots. In addition, one-way PERMANOVA confirmed a significant difference in microbial communities between these two groups at the phylum ($p = 0.03$), family ($p = 0.05$), and genus ($p = 0.04$) levels (see Supplementary Table 12, Fig. 3).

Comparative analysis of urinary Microbiome composition in healthy cats and cats with feline idiopathic cystitis

When comparing cats with FIC to healthy cats, a total of 3 phyla, 5 classes, 10 orders, 12 families, and 12 genera were identified (see Supplementary Table 13, Fig. 4). The phylum Proteobacteria was the most prevalent

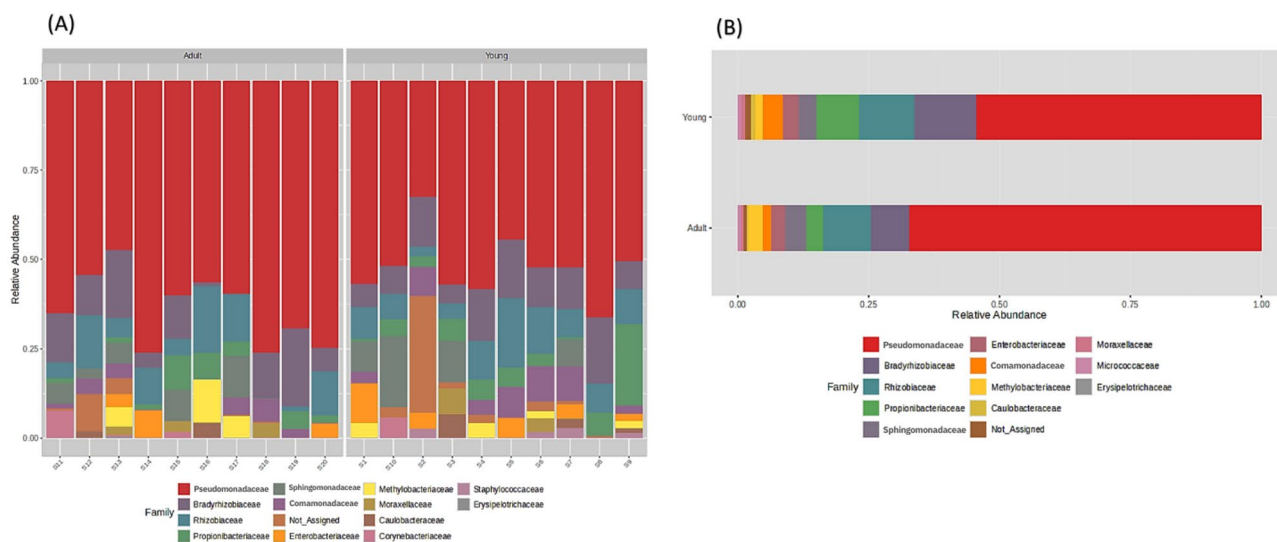


Fig. 1. Relative abundance of bacterial families in the urine of healthy young and adult cats. (A) displays the bacterial family abundance in urine samples from 10 young and 10 adult cats, with *Pseudomonadaceae* identified as the predominant family across all samples. (B) shows the differences in bacterial family abundance between the young and adult groups, demonstrating that *Pseudomonadaceae* remains the dominant family in both age groups, though its relative abundance is significantly lower in young cats compared to adults ($p = 0.02$).

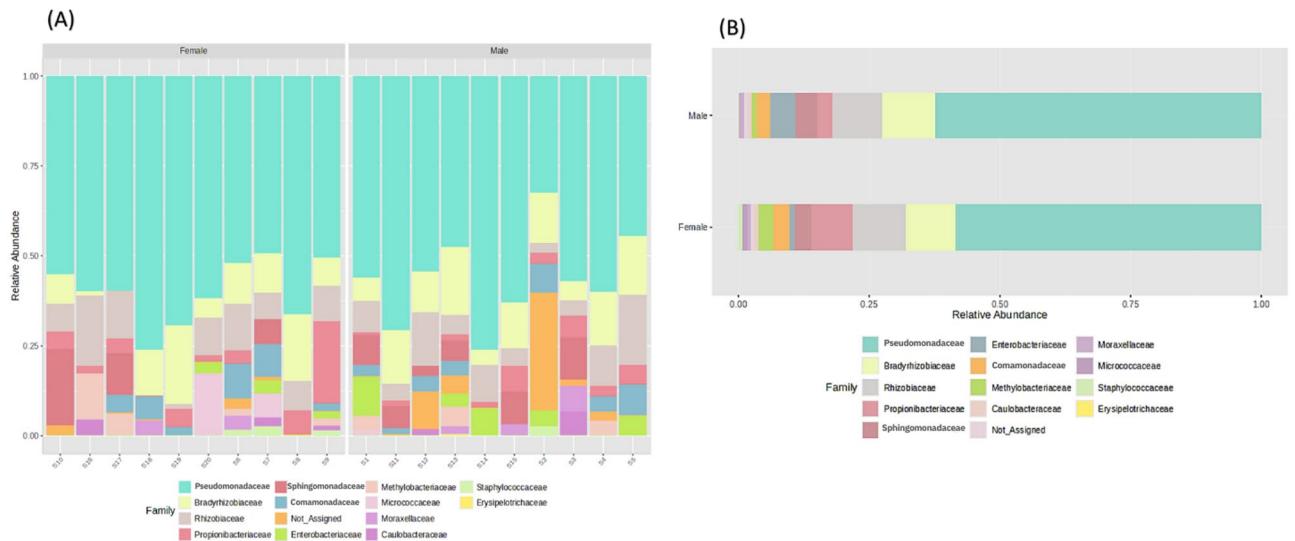


Fig. 2. Relative abundance of bacterial families in the urine of 10 male and 10 female cats. (A) presents the bacterial family composition for each individual, with *Pseudomonadaceae* consistently identified as the predominant family. (B) compares family-level abundances between male and female groups, showing that *Pseudomonadaceae* remained the dominant family in both sexes, with no statistically significant differences in its relative abundance between the groups.

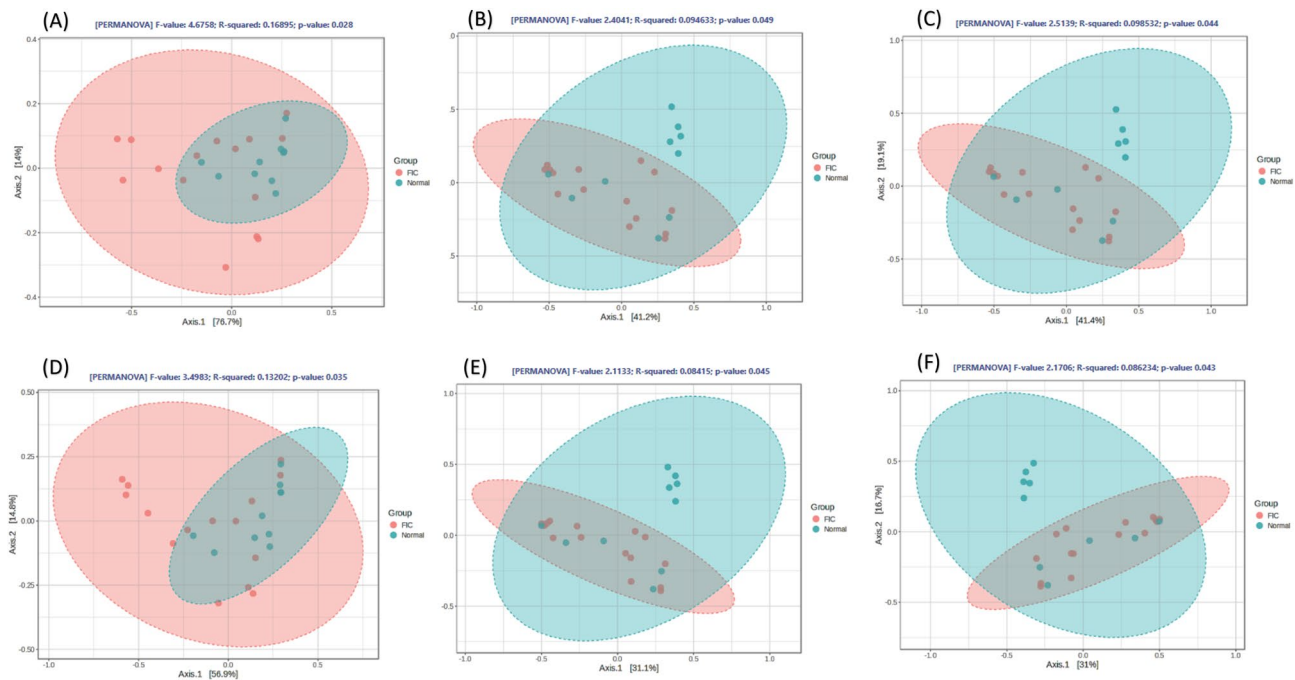


Fig. 3. Principal Coordinate Analysis (PcoA) plot of beta diversity, as measured by Bray-Curtis at (A) the phylum level, (B) the family level, and (C) the genus level, as well as the Jaccard index at (D) the phylum level, (E) the family level, and (F) the genus level, revealed significant different in bacterial communities between cats with feline idiopathic cystitis (FIC) and normal cats.

in the normal group, with a significantly higher abundance in normal cats (81.0%) compared to FIC cats (45.3%) ($p=0.02$). In contrast, the phylum Firmicutes were significantly more prevalent in FIC cats (47.0%) compared to normal cats (10.0%) ($p=0.03$). Similarly, the class Bacilli, which belongs to the phylum Firmicutes, was significantly more prevalent in FIC cats (40.7%) compared to healthy cats (10.0%) ($p=0.03$). The genus *Corynebacterium* showed a significantly reduced relative abundance in cats with FIC (0.25%) compared to normal controls (2.35%) ($p=0.05$), with no significant differences observed at the phylum and class levels.

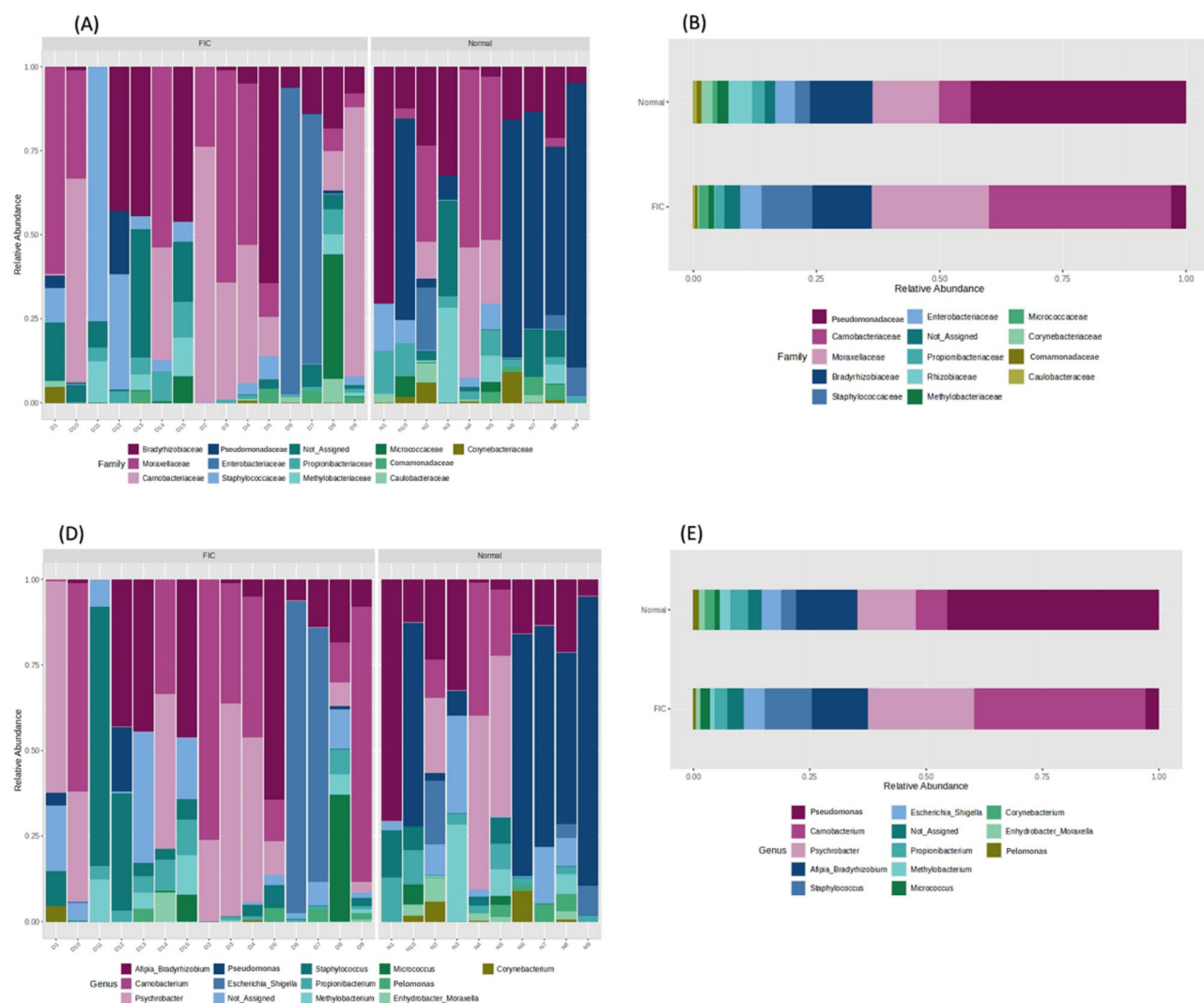


Fig. 4. Relative abundance of bacterial families and genera in the urine of 10 healthy cats and 15 cats diagnosed with feline idiopathic cystitis (FIC). (A) shows the family-level bacterial composition for each cat, while (B) compares family-level abundances between the healthy and FIC groups. (C) presents the genus-level composition for each cat, and (D) compares genus-level abundances between the groups. Notably, cats with FIC exhibited reduced abundance of the family *Pseudomonadaceae* and the genus *Pseudomonas* compared to healthy cats.

Additionally, the genus *Pseudomonas* belonging to the family *Pseudomonadaceae* in the phylum Proteobacteria was significantly higher abundance in normal cats (45.3%) compared to FIC cats (2.9%) ($p=0.01$).

Discussion

This study investigated the bacterial profiles in the urine of healthy cats of different ages and sexes, and compared them with those of cats suffering from idiopathic cystitis using sequence analysis of the 16S rRNA gene marker, specifically the V3–V4 hypervariable regions. When interpreting these findings, it is essential to recognize that the feline urinary microbiome is likely influenced by multiple interacting factors, including host characteristics, methodological variables (such as sample collection and laboratory procedures), and environmental factors (such as diet, medication, husbandry practices, and geographic location). These factors may contribute to variability within and between investigations, highlighting the challenges in comparing urinary microbiome findings across studies.

Consistent with previous studies that reported a limited presence of bacterial taxa in the urine of cats^{8,10,12}, our findings revealed a low number of bacterial taxa. It is generally observed that urine contains lower quantities of bacterial DNA compared to other body sites, such as the gut, oral cavity, or skin, resulting in a less abundant and less diverse microbial population detected²³. Fewer bacterial taxa are generally found in the urine of cats compared to microbiome studies of human and dog urine^{24,25}. Several factors may influence the variation in the number of bacterial taxa, including the different genetic and environmental backgrounds of the studied populations, which could act as confounding factors. Additionally, the method of sampling is another factor

that affects the abundance and composition of bacterial communities in both humans and animals^{26,27}. Voided or catheterized techniques are utilized in human studies²⁸, whereas cystocentesis was commonly used in animal studies^{10,24}, including our investigation. It has been observed that urine collected via voiding in dogs contains a greater abundance of taxa than urine extracted through cystocentesis²⁶. Using voided or catheterized techniques can introduce bacteria from sources outside the bladder, particularly the genital tract and skin, leading to an increased bacterial population from adjacent tissues.

Interestingly, certain bacterial taxa identified in our study were consistent with previous investigations based on 16S rRNA sequencing^{8,10}. The majority of the taxa present in healthy cats belonged to the phylum Proteobacteria. Furthermore, numerous bacterial taxa identified were similar to those reported in the urine of healthy cats. These taxa include bacteria from the orders Pseudomonadales, Rhizobiales, Sphingomonadales, Enterobacteriales, Propionibacteriales, Bacillales, and Micrococcales. Therefore, these findings support the idea that the urinary bladder of healthy cats harbors a resident microbiota, and these taxa may constitute part of the bacterial structure in cats' urine. Notably, there are several uropathogenic genera within the Proteobacteria phylum, including *Escherichia* spp., *Bordetella* spp., *Proteus* spp., *Klebsiella* spp., *Pasteurella* spp., *Haemophilus* spp., and *Pseudomonas* spp., which was detected in our study. Thus, alterations in commensal bacteria may potentially contribute to urinary tract pathology.

In the age-based comparison between young cats (< 1 year, $n = 10$) and adult cats (> 1 year, $n = 10$), young cats demonstrated significantly higher alpha diversity across multiple taxonomic levels, indicating greater richness and evenness in their urinary microbiota. However, beta diversity analyses using both Bray–Curtis and Jaccard indices revealed no significant differences in overall community structure between the age groups. This suggests that age-related changes in healthy cats may reflect a loss of within-sample microbial diversity rather than a major compositional shift, in accordance with findings in other low-biomass microbiomes, where reductions in a small subset of taxa can significantly affect alpha but not beta diversity²⁹. Supporting this observation, several predominant taxa were more abundant in adult cats, whereas certain bacterial groups, including Phylum Firmicutes, Class Bacilli, Order Bacillales, and Family *Staphylococcaceae*, were exclusively detected in the young group. Notably, these taxa were present in only 4 of the 10 young cats, underscoring their lack of universal presence and inability to serve as an age-defining markers. Given the preliminary nature of our findings, caution is warranted when interpreting taxa-level differences due to the small sample size, which increases susceptibility to individual variation and stochastic detection—an acknowledged challenge in urinary and other low-biomass microbiome studies^{23,30}. In humans, similar age-related changes in the urinary microbiome have been documented^{13,14,31}, with studies indicating that the core urinary microbiome fluctuates in abundance between age groups, and certain genera are detected exclusively in older individuals. In cats, larger cohort studies with more statistical power are essential to confirm whether these taxa represent meaningful age-associated features of the feline urinary microbiome, particularly in light of previous reports demonstrating inconsistent taxa detection across studies of feline urine.

In the comparison of genders, the urine microbiota of male and female cats exhibited similar composition and abundance. Furthermore, no differences were observed between the two bacterial communities in terms of beta diversity. This finding aligns with a previous study on dogs' urine, which reported no significant differences in the urinary microbiome between male and female dogs²⁴. Additionally, despite the use of different urine collection techniques, no discernible sex differences were detected in the urinary microbiome of dogs²⁶. In contrast, findings in humans indicate that while the urinary microbiome of females and males comprises similar taxa, there are notable differences between healthy males and females^{28,32}. Specifically, females tend to exhibit a more heterogeneous bacterial population than males. The core microbiome in both men and women typically includes *Lactobacillus*, *Corynebacterium*, and *Streptococcus*. *Corynebacterium* and *Streptococcus* were found to be predominant in the urinary microbiome of healthy men^{33,34}, while *Lactobacillus* was the most dominant genus in the urinary microbiome of healthy women^{35–38}. Additionally, hormonal effects have been documented to influence the urine microbiome in females during the menstrual cycle³⁹, which could potentially explain the observed differences in urinary microbiomes between genders and among individual females. The origin of the urinary microbial community remains unclear. A study has shown similarities between the urine microbiota of women and their vaginal microbiota⁴⁰. Therefore, the differences in urine microbial composition between men and women may be associated with distinct resident bacteria present in the genitals of each gender.

In cats, several bacterial genera have been identified as normal flora in the genital tract of healthy individuals through culture isolation⁴¹. Our study also identified certain bacteria from the families *Staphylococcaceae*, *Moraxellaceae*, *Enterobacteriaceae*, and *Corynebacteriaceae*, consistent with the findings of Ström Holst et al. (2003)⁴¹. This overlap has led to speculation that the lower genital tract may contribute to the urinary microbiome. Although the genital microbiota differs between sexes and is influenced by factors such as the estrous cycle^{41,42}, we observed no sex-associated differences in urinary microbiome composition. This finding implies that the bladder microbiome may be relatively stable and distinct from genital tract flora. Additionally, the methodology employed—specifically cystocentesis—effectively bypasses the urethra, likely minimizing contributions from genital-associated bacteria. Therefore, while some taxa may be shared between the genital and urinary tracts, our results suggest that the bladder microbiome likely possesses an independent origin or ecological maintenance mechanism that is not exclusively related to sex-based differences in genital flora. Further studies incorporating paired urine and genital tract sampling across various hormonal states are essential to clarify the biological origins and stability of the feline urinary microbiome.

The comparison of urinary microbiota between healthy cats and those with feline idiopathic cystitis (FIC) revealed differences between the two groups. Our findings indicate that the majority of bacteria identified in all cats in this study belong to the order Pseudomonadales. Interestingly, there was a significant reduction in the *Pseudomonadaceae* and *Corynebacteriaceae* families, as well as the *Pseudomonas* and *Corynebacterium* genera, in the urine of cats with FIC compared to normal cats. These findings differ from those of two previous studies:

Balboni et al. (2020) reported that the order Clostridiales was notably abundant in normal cats⁸, while Kim et al. (2021) demonstrated the prominence of the *Escherichia-Shigella* genus, which belongs to the *Enterobacteriaceae*, in the urine of normal cats¹⁰. Furthermore, the latter study noted the presence of the *Pseudomonas* genus in normal cats, although it was less abundant than the *Escherichia-Shigella* genus¹⁰. However, they found no significant differences in bacterial composition between cats with FIC and normal cats. In humans, significant differences in urinary microbiota, notably a decrease in *Lactobacillus* and an increase in opportunistic pathogens were found in patients with interstitial cystitis^{19,43}. However, the underlying reasons for the increase in opportunistic pathogens and whether the loss of microbial diversity is a cause, or a result of the interstitial cystitis condition remain unclear. Notably, the median age of cats with FIC was 3.0 years, compared to 2.0 years for healthy controls; however, this difference was not statistically significant. Nonetheless, age variation between groups may represent a potential confounding factor influencing urinary microbial profiles. A previous study has shown that age can affect microbial diversity and host immune responses⁴⁴, which may partially contribute to observed differences between groups. Therefore, future studies with age-matched cohorts or models that adjust for age are warranted.

There are several potential factors that may contribute to the discrepancies observed in the dominant bacterial taxa between our findings and those reported in previous studies^{8,10,11}. Both methodological⁴⁵ and biological factors^{44,46} are likely to play major roles in driving variation in the urinary microbiome of cats. Variability in sample collection methods, timing, and handling can influence microbial detection^{26,47}, particularly in low-biomass specimens like urine. Differences in DNA extraction protocols may also introduce analytical bias. Furthermore, the choice of primer sets, sequencing depth, and library preparation can selectively amplify or underrepresent specific taxa^{48,49}. Downstream bioinformatic pipelines, including database selection, taxonomic assignment algorithms, and filtering thresholds, can yield divergent results even when analyzing identical raw data. Lastly, intrinsic biological variation among cats such as age, diet, prior antimicrobial exposure, stress levels, and urine concentration, along with geographic location and husbandry practices across different cat populations, may contribute to additional heterogeneity and influence the urinary microbiome, further amplifying inter-study variability^{14,46,50,51}.

Based on our findings, it is not possible to draw definitive conclusions about how alterations in microbiota are associated with the pathophysiology of FIC. Changes in the microbiome may occur from two perspectives. First, changes in microbial communities may induce inflammation of the bladder. It is now understood that idiopathic cystitis is typically recognized as a type of bladder inflammation that is not caused by an infection^{52,53}. However, this issue remains complex and unclear, as bovine herpes virus type 4, feline calicivirus, and feline foamy virus have all been implicated as potential causes of FIC^{54,55}. In addition, the undergrowth or overgrowth of certain microbes may have the capacity to secrete or produce biological substances and metabolites that can stimulate the urothelium and trigger an inflammatory response. Changes in these microbial components could potentially contribute to the development of bladder inflammation in FIC. Secondly, bladder inflammation may initiate changes in the bladder environment, which subsequently induce alterations in the microbiota. A previous study detected various metabolites in the urine of cats with FIC, including carbonyl, peptide, and carbohydrate molecules²². Both inflammation and the differences in urinary metabolites have the potential to impact fluctuations in the urine environment, which may promote or inhibit the growth of specific microorganisms, leading to variations in the abundance and diversity of bacteria^{56,57}. However, relying solely on 16S rRNA amplicon sequencing does not allow us to determine the cause of the observed differences in bacterial composition, whether alterations in the urinary microbiota play a role in the pathogenesis of idiopathic cystitis, or if these changes are secondary effects associated with FIC. In addition, 16S rRNA sequencing cannot distinguish viable bacteria from non-viable cells, which means that the microbial signatures detected may partly reflect residual bacterial DNA rather than metabolically active communities. Therefore, a comprehensive evaluation of the urinary microbiome and metabolome, combined with the identification of metabolic pathways involved in microbe-metabolite-host interactions in cats with FIC, is warranted in future studies.

There are limitations in our study. DNA sequencing-based methods are unable to differentiate between live and dead bacteria, and to determine the functional properties of the identified bacterial species. As a result, the findings of this study cannot confirm the viability of the bacteria or elucidate their roles in relation to the host. A combination of culture-based approaches, such as expanded quantitative urine culture (EQUC), which employs larger urine volumes and different growth media than standard protocols, along with NGS, could provide more comprehensive insights and is worth exploring. Another limitation is that the taxonomic resolution of short-read sequencing, when using Illumina MiSeq amplicon sequencing, is limited to the genus level, which may restrict the depth of analysis. Additionally, the classification of study subjects may have introduced confounding factors. Since serum thyroid hormone levels were not measured, the possibility of undiagnosed hyperthyroidism in some of the enrolled cats cannot be excluded. Hyperthyroidism is a well-recognized cause of polyuria and lower urinary tract symptoms in older cats, which may mimic or influence clinical presentations similar to FIC⁵⁸. However, current literature provides limited information regarding the urinary microbiome in hyperthyroid cats, making it difficult to predict how this condition might have influenced the microbiome profiles observed in this study. Furthermore, the presence of urethral strictures in cats diagnosed with FIC could not be definitively ruled out, as diagnostic evaluation was limited to abdominal ultrasonography and plain radiography; advanced imaging techniques such as cystourethrography, contrast urethrography, or computed tomography (CT) were not performed. Additionally, since the cats in this study were client-owned and enrolled at an animal hospital, it was not possible to control the type of food and drinking water provided to them. Dietary supplements, particularly probiotics, can vary widely among different brands of food and snacks, potentially altering the urinary microbiome. Dietary variations may influence urinary microbiomes by modifying urine metabolites, resulting in distinct bacterial abundances and diversity^{50,51,56,57}. Similarly, the types of cat litter boxes and hygiene practices of cat owners could not be standardized, which may further contribute to variability in the

urinary microbiome among individual cats. Future studies with larger sample sizes that account for potential confounding factors could help to corroborate the findings of this study.

Conclusion

Our findings indicate that cat urine contains diverse microbial communities, with the urinary microbiota of healthy cats being predominantly composed of the Proteobacteria phylum, particularly the *Pseudomonas* genus. While no significant differences were observed between sexes, minor variations were noted between age groups. In cats with FIC, there was a decrease in the Proteobacteria phylum, accompanied by an increase in the Firmicutes phylum and the Bacilli class, as well as a decrease in the *Pseudomonas* and *Corynebacterium* genera. Although the precise role of the microbiome in the bladder remains unclear, this research enhances our understanding of feline urinary microbiota and provides a foundation for future studies. Key areas for further exploration include identifying microbial signatures as potential diagnostic biomarkers for FIC and employing multi-omics approaches to elucidate the microbiome's role in the pathophysiology of FIC.

Materials and methods

Animals

This prospective study was conducted on client-owned cats enrolled at the Veterinary Teaching Hospital of the Faculty of Veterinary Medicine, Khon Kaen University, Thailand, from 2021 to 2022. All animal procedures complied with the ethical principles and guidelines for the use of animals in scientific research established by the National Research Council of Thailand, with prior informed consent obtained from the cat owners. The study was approved by the Supervision and Monitoring on Animal Care and Use Center of Khon Kaen University (approval numbers: 47/64 and 142/66) and reported in accordance with ARRIVE guidelines (<http://arriveguidelines.org>).

Healthy cats were defined based on an evaluation of their health status, which included the absence of recent disease history, clinical signs, and abnormal clinicopathological findings. To assess the health status of each cat, routine blood samples were collected for hematology, serum chemistry profiling, and testing for FeLV/FIV status (Witness[®], Zoetis, New Jersey, USA). Additionally, abdominal ultrasound, urinalysis, and urine culture were performed on all cats. Animals were excluded from the study if bacterial growth was detected in aerobic urine cultures, or if the hematological, serum biochemical, urinalysis, or abdominal ultrasound results were abnormal. Furthermore, animals that had received antibiotics or probiotics within the previous 30 days were also excluded.

Cats with feline idiopathic cystitis (FIC) were defined as those presenting clinical signs of lower urinary tract disease, such as hematuria, dysuria, stranguria, pollakiuria, or periuria. The diagnosis of FIC was based on clinical signs, physical examination, urinalysis, aerobic urine culture, abdominal ultrasound, and/or abdominal radiographs. Cats with FLUTD were excluded from this study if: (i) they had been diagnosed with diseases other than IC; (ii) they had a positive urine culture; (iii) they showed evidence of systemic infection; or (iv) they had received antibiotics, anti-inflammatory drugs, and/or probiotics within the previous 30 days.

To characterize bacterial communities in the urine of healthy cats of different ages and genders, we recruited both male and female cats aged 6 months to 5 years. Cats aged 1 year or older were classified as adults, while those younger than 1 year were considered young cats⁵⁹. Twenty cats were enrolled in the study and divided into four groups, each consisting of 5 cats: young males, young females, adult males, and adult females.

To compare the bacterial communities in the urine of healthy cats and those with FIC, male cats aged ≥ 1 year, whether healthy or with IC, were included. The enrolled cats were divided into two groups: the control group, consisting of healthy cats ($n = 10$), and the disease group, composed of cats with IC ($n = 15$).

Blood collection and analysis

Blood and urine collection were performed on all enrolled cats for health checkup. A total of 1 ml of blood was collected by cephalic or saphenous venipuncture, with 0.5 ml placed in an EDTA tube for hematology analysis and the remaining 0.5 ml transferred into a lithium heparin tube for biochemistry profiles, including BUN, creatinine, ALT, ALP, total protein, albumin, globulin, phosphorus, amylase, glucose, sodium, and potassium. Hematology analysis was conducted using BC-5000 Auto Hematology analyzer (Mindray, Shenzhen, China), and serum biochemistry was analyzed by a Vetscan[®] VS2 analyzer (Zoetis, New Jersey, USA).

Collection of urine samples

Urine samples were collected by cystocentesis. Some cats received a light sedation of acepromazine 0.02 mg/kg, intramuscular route to minimize stress and any discomfort before cystocentesis, as suggested by Robertson et al. (2018)⁶⁰. Cats were positioned in dorsal recumbency, and then the fur at the bladder site was shaved and disinfected with surgical cleaning agents to prevent contamination before urine collection. The skin scrub was performed using 4% chlorhexidine gluconate rinsed with 70% isopropyl alcohol. This procedure was repeated three times. A total of 6 ml of urine was collected using ultrasound guidance with a 24G x 1-inch sterile needle connected to a 10 ml sterile syringe, and the sample was transferred to a sterile tube. One milliliter of urine was used for aerobic bacterial culture and urinalysis, while 5 ml was mixed with 350 μ l of Urine Conditioning Buffer (Zymo Research, Irvine, CA) for DNA extraction and 16S rRNA sequencing.

Urinalysis and aerobic bacterial culture

Standard urinalysis was performed on all samples immediately, which included analysis using UriSCAN test strip (YD Diagnostics, Gyeonggi-do, Korea), urine specific gravity measurement with ATAGO[®] PAL-CAT refractometer (Atago, Kyoto, Japan), and microscopic examination of sediment. Aerobic bacterial culture of urine samples was processed within 30 min after urine collection. Urine samples of 1 μ l, 10 μ l and 100 μ l were

inoculated onto blood agar and MacConkey agar plates which were then incubated at 37 °C under aerobic conditions. Plates were inspected for colony formation after 24 and 48 h. Any growth observed indicated bacteriuria, leading to the exclusion of the animals from the study.

16S rRNA sequencing

Urine sample preparation was performed following the Zymo Research's protocol. The urine was centrifuged at 3,000 $\times$ g for 15 min, and the supernatants were discarded. The pellets were resuspended in 1 ml of DNA/RNA shield (Zymo Research, Irvine, CA) to stabilize nucleic acids. The samples were then stored at –20 °C until shipping for DNA extraction, library preparation and sequencing process at Zymo Research Laboratory, California, USA. Briefly, total bacterial DNA was extracted from each urine sample using ZymoBIOMIC[®] –96 MagBead DNA kit (Zymo Research, Irvine, CA) according to the manufacturer's instructions. For library preparation, the extracted samples were processed with *Quick-16S*[™] NGS Library Prep Kit (Zymo Research, Irvine, CA). In the step of 16S rRNA gene amplification, the *Quick-16S*[™] Primer Set V3-V4 (Zymo Research, Irvine, CA), designed for optimal coverage of the 16S gene, was used in the PCR reactions. The final PCR products were quantified using qPCR fluorescence readings and pooled together based on equal molarity. The pooled DNA library was cleaned up with the Select-a-Size DNA Clean & Concentration[™] (Zymo Research, Irvine, CA), and quantified using TapeStation[®] (Agilent Technologies, Santa Clara, CA) and Qubit[®] (Thermo Fisher Scientific, Waltham, WA).

The ZymoBIOMICS[®] Microbial Community Standard (Zymo Research, Irvine, CA) was used as a positive control for each DNA extraction, and ZymoBIOMICS[®] Microbial Community DNA Standard (Zymo Research, Irvine, CA) was used as a positive control for each targeted library preparation. Negative controls, including blank extraction control and blank library preparation controls, were included to assess the level of bioburden carried by the wet-lab process. The final DNA library was sequenced on the Illumina[®] Miseq[™] using the v3 reagent kit (600 cycles), incorporating a 10% PhiX spike-in. Raw reads were the final output of the sequencing process.

Bioinformatics analysis

The raw reads represent individual DNA fragments that have been sequenced. These raw reads underwent processing and analysis, including filtering, trimming, merging paired reads, chimera removal, assembly, and annotation using Quantitative Insight Into Microbial Ecology (QIIME Version 1.9.1) software (<https://qiime.org/install/install.html>). The sequences were processed following the DADA2 v1.16.0 pipeline, available at <https://benjjneb.github.io/dada2/>. The DADA2 pipeline describes microbial diversity and community structures using unique amplicon sequence variants (ASVs). Microbial Taxonomy was assigned using the Zymo Research Database as a reference database. The compiled data were used to determine the relative percentage of bacterial composition for each individual sample. Composition visualization, alpha diversity (observed number, Chao1, Shannon, and Simpson) and beta diversity analysis (Bray-Curtis and Jaccard Index) were computed using Microbiome Analyst version 2.0 (www.microbiomeanalyst.ca), an online platform for microbiome dataset analysis, visualization, interpretation, and comprehensive statistics^{61–63}.

Statistical analysis

Descriptive statistics were used to summarize demographic variables, including age and sex. Data following a normal distribution were reported as mean $\pm$ standard deviation (SD), whereas data not normally distributed were presented as median and interquartile range. Microbiome profiling, including alpha and beta diversity, as well as taxonomic composition analysis, was conducted using MicrobiomeAnalyst version 2.0 (<https://www.microbiomeanalyst.ca>), which integrates multiple R-based statistical packages as its analytical backend^{61–63}. The Mann-Whitney U test was used to assess differences between groups of samples, in terms of alpha diversity, and relative abundance. Permutational Multivariate Analysis of Variance (PERMANOVA) was performed to evaluate significant differences in beta diversity between groups.

Data availability

The data generated in this study are included within the manuscript and supplementary materials and can be provided upon request. DNA sequences data generated during the study have been deposited with links to BioProject accession number PRJNA1263522 in the NCBI BioProject database (<https://www.ncbi.nlm.nih.gov/bioproject/1263522>).

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Author contributions

P.T. investigation, data curation, formal analysis, methodology, validation, writing-original draft preparation. N.F. supervision, conceptualization, formal analysis, methodology, validation, writing-review and editing. P.S. methodology, statistical analysis, writing-review and editing. P.P. conceptualization, investigation, methodology, project administration, supervision, validation, writing-original draft preparation, review and editing. All the authors have read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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